



PAY

Digital Payment Systems (DPS) in India

By

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Digital Financial Payment Systems in India

- **India has witnessed a significant rise in digital payments in recent years, driven by government initiatives, technological advancements, and changing consumer behavior.**
- **This lecture will explore the landscape of digital payments in India, highlighting its benefits, types of systems, and future trends.**



The Importance of Digital Financial Literacy in India for Sustainable Growth:

- India is a developing economy with a focus on sustainable growth, and digital financial literacy is a pressing issue.
- To achieve sustainable development, it's essential that every section of society has access to equal opportunities to participate in nation-building.
- The lack of digital financial literacy among the rural population is a significant challenge, particularly given the Government's recent demonetization and plans to transition to a cashless economy.
- Therefore, there's a critical need to increase awareness and facilitate access to digital financial services, especially in rural and semi-urban areas. Educating and empowering citizens to leverage digital financial services is an urgent need.

Digital Financial Inclusion and Digital Financial Services (DFS):

- **Digital financial inclusion** is the act of providing excluded and marginalized groups of society with digital access to formal financial services.
- This is accomplished through the implementation of **Digital Financial Services (DFS)**, which are tailored to meet customers' requirements and delivered in a responsible manner at a cost that is both sustainable for providers and affordable for customers.
- The three main components of DFS are:
 - a digital transactional platform,
 - retail agents, and
 - the use of a device, typically a mobile phone, by customers and agents to conduct transactions through the platform.

Digital Financial Inclusion and Digital Financial Services (DFS):

- Increasingly, the **unbanked population** is gaining access to financial services through digital channels, which has become a significant means for the same.
- In this regard, banks, microfinance institutions, mobile operators, and third-party providers have leveraged mobile phones, point-of-sale (POS) devices, and networks of small-scale agents to offer basic financial services.
- These digital channels offer greater convenience, scalability, and cost-effectiveness than traditional banking methods.



Financial Inclusion Schemes by Gol:

- **Jan Dhan-Aadhar-Mobile (JAM) Trinity:**
 - The combination of Aadhaar, PMJDY, and a surge in mobile communication has reshaped the way citizens access government services.
 - By significantly changing the concept of individual identity, Aadhaar has not only brought about a secure and easily verifiable system but also easy to obtain as well to help in the financial inclusion process.
 - The government has also launched many flagship schemes to promote financial inclusion and provide financial security to empower the poor and unbanked in the country. These include:
 - the Pradhan Mantri Mudra Yojana,
 - Stand-Up India Scheme,
 - Pradhan Mantri Jeevan Jyoti Bima Yojana,
 - Pradhan Mantri Suraksha Bima Yojana, and
 - Atal Pension Yojana.





World's Largest Financial Inclusion Scheme

No charges for opening & maintaining of account

No minimum balance required to be maintained in account

Free RuPay debit card

Accidental insurance cover Rs. 2 lakh (Rs. 1 lakh for cards issued to accounts opened upto 28.08.2018) on RuPay card

Interest on deposit



Access to pension and micro-insurance products

Beneficiaries of Government schemes can get DBT in these accounts

Facility of making digital payments

Easy transfer of money across India

OD upto Rs. 10,000 to eligible account holders

Types of Digital Payment Systems in India:

Types of Digital Payment Systems in India:

- i. **Debit and Credit Cards**: Traditional payment methods accepted by merchants worldwide.
- ii. **Unified Payments Interface (UPI)**: A real-time interbank payment system that allows users to transfer money instantly using a mobile app and a virtual payment address (VPA).
- iii. **Bharat Interface for Money (BHIM)**: A mobile app that facilitates payments using UPI and other methods like Aadhaar-enabled payments.
- iv. **Mobile Wallets**: Electronic wallets stored on mobile devices that allow users to pay for goods and services online or through contactless payments.
- v. **Aadhaar Enabled Payment System (AEPS)**: Enables bank account holders to perform financial transactions using their Aadhaar number and fingerprint authentication.
- vi. **Internet Banking**: Online banking platforms that allow users to access their accounts and make payments.
- vii. **Bharat Bill Payment System (BBPS)**: A platform for consolidated bill payments across different categories.



Credit / Debit Cards:

- **Credit card:** A line of credit issued by a bank, allowing you to borrow money to make purchases. You must pay back the borrowed amount with interest.
- **Debit card:** Directly linked to your checking account. When you use it, the money is deducted immediately from your account balance.



Credit Cards:



- **Benefits:**

- **Building credit history:** Responsible credit card use helps build a positive credit history, which can improve your credit score and make it easier to qualify for loans and other financial products.
- **Rewards and benefits:** Many credit cards offer loyalty programs, cashback, travel benefits, and purchase protection.
- **Convenience:** Provides a backup option if you don't have enough cash on hand.
- **Improved budgeting:** Track your spending through monthly statements.

- **Fees:**

- Annual fees
- Interest charges on unpaid balances
- Late payment fees
- Foreign transaction fees

Debit Cards:

- **Benefits:**
 - **Spend only what you have:** No risk of accruing debt or interest charges.
 - **Reduced impulse spending:** Helps control your budget by limiting spending to your available balance.
 - **Immediate access to funds:** No need to wait for funds to clear, as with checks.
 - **Widely accepted:** Most merchants accept debit cards.
- **Fees:**
 - ATM withdrawal fees
 - Overdraft fees
 - Monthly service fees on some debit cards





Credit or Debit Card – Choosing the right Card:

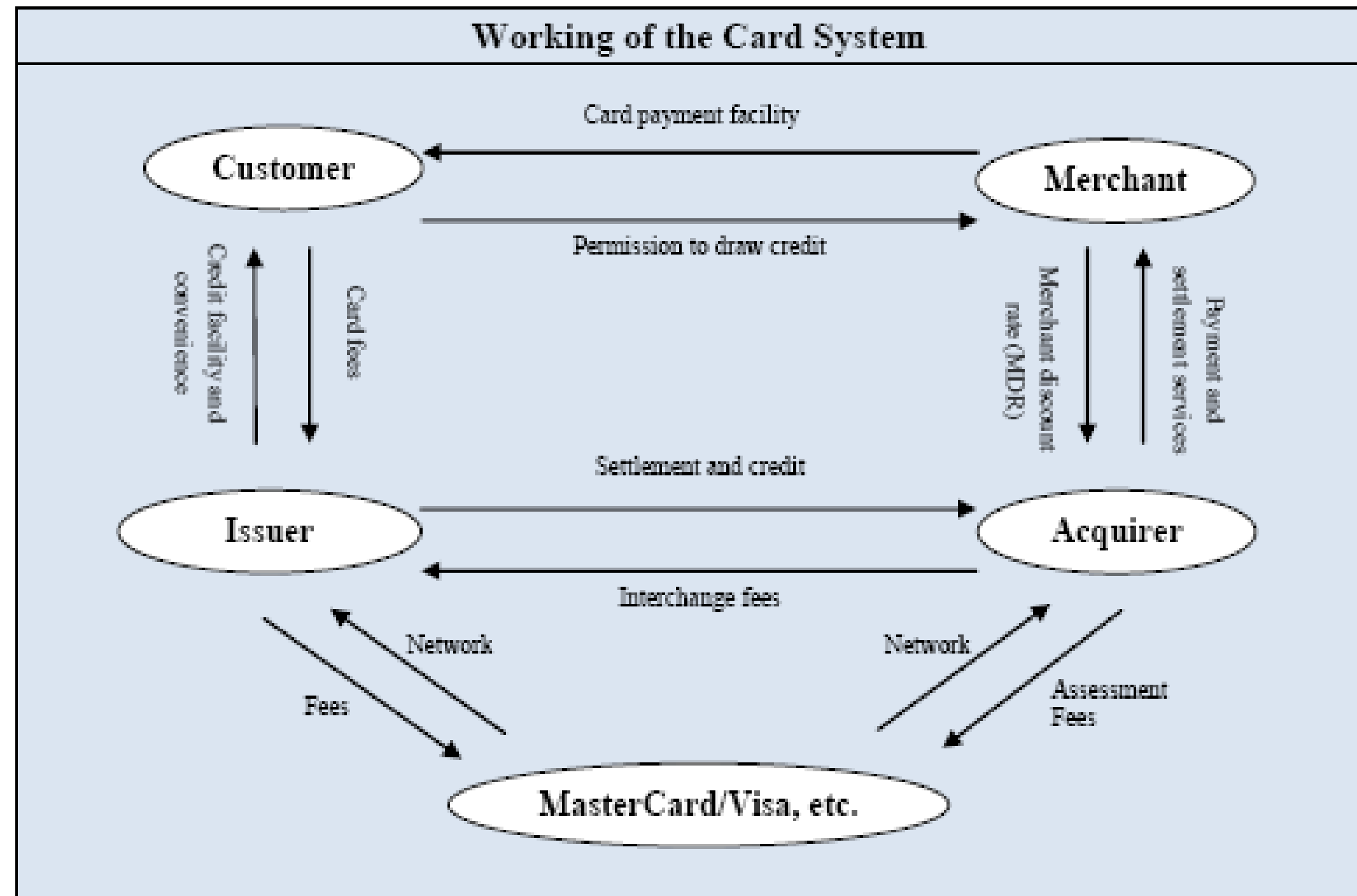
- **The best choice depends on your individual needs and spending habits. Consider factors like::**
 - **Creditworthiness:** If you have poor credit, it may be difficult to qualify for a good credit card.
 - **Spending habits:** If you tend to overspend, a debit card may be a better option to avoid debt.
 - **Financial goals:** Consider rewards programs and benefits that align with your goals.

Working of Card Based Payment Systems

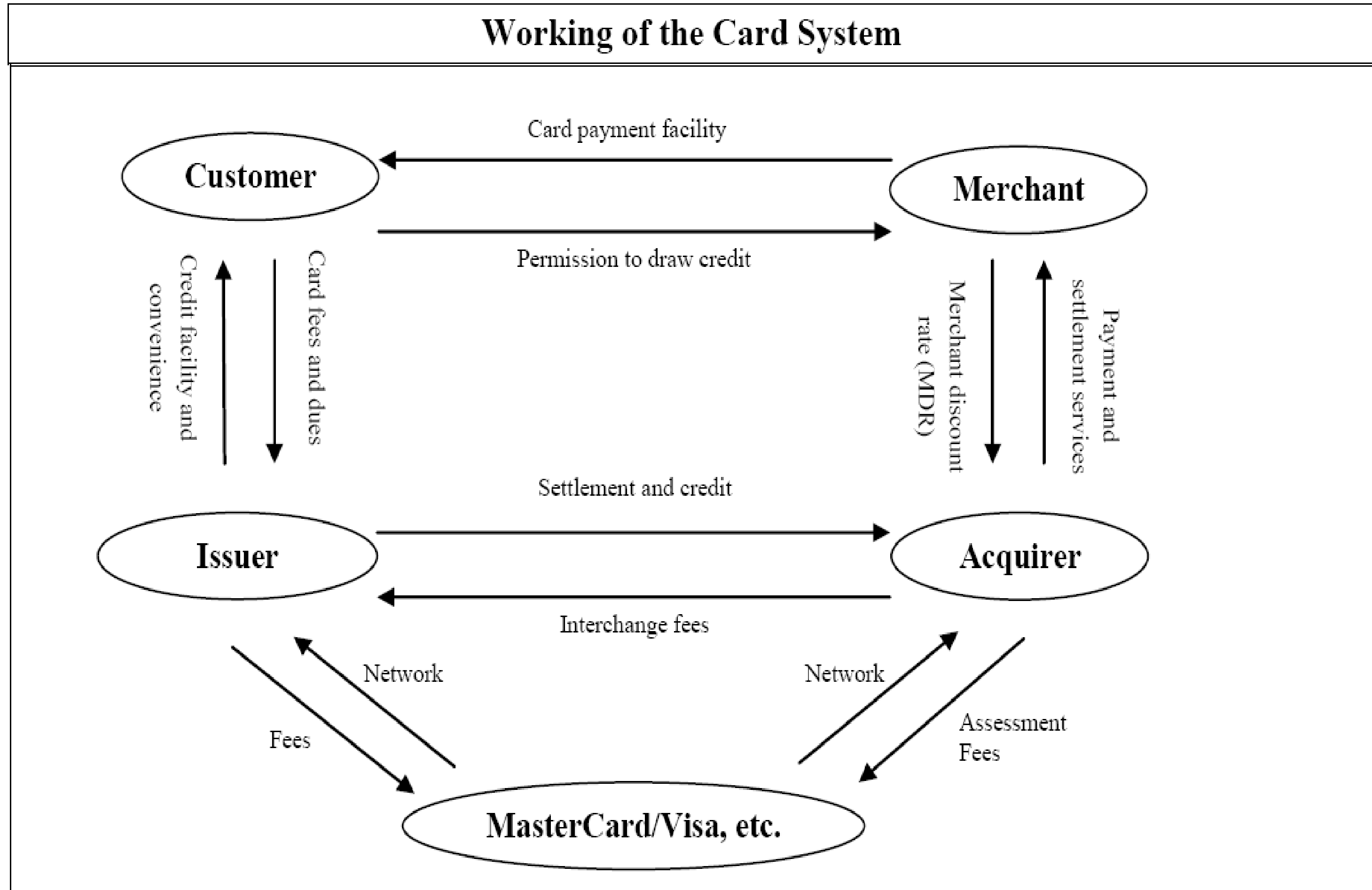
- The card based payment system functions with its players that include:
 - The card companies (switch provider)
 - Banks (acquirer and issuer)
 - Merchants
 - Cardholders



POS

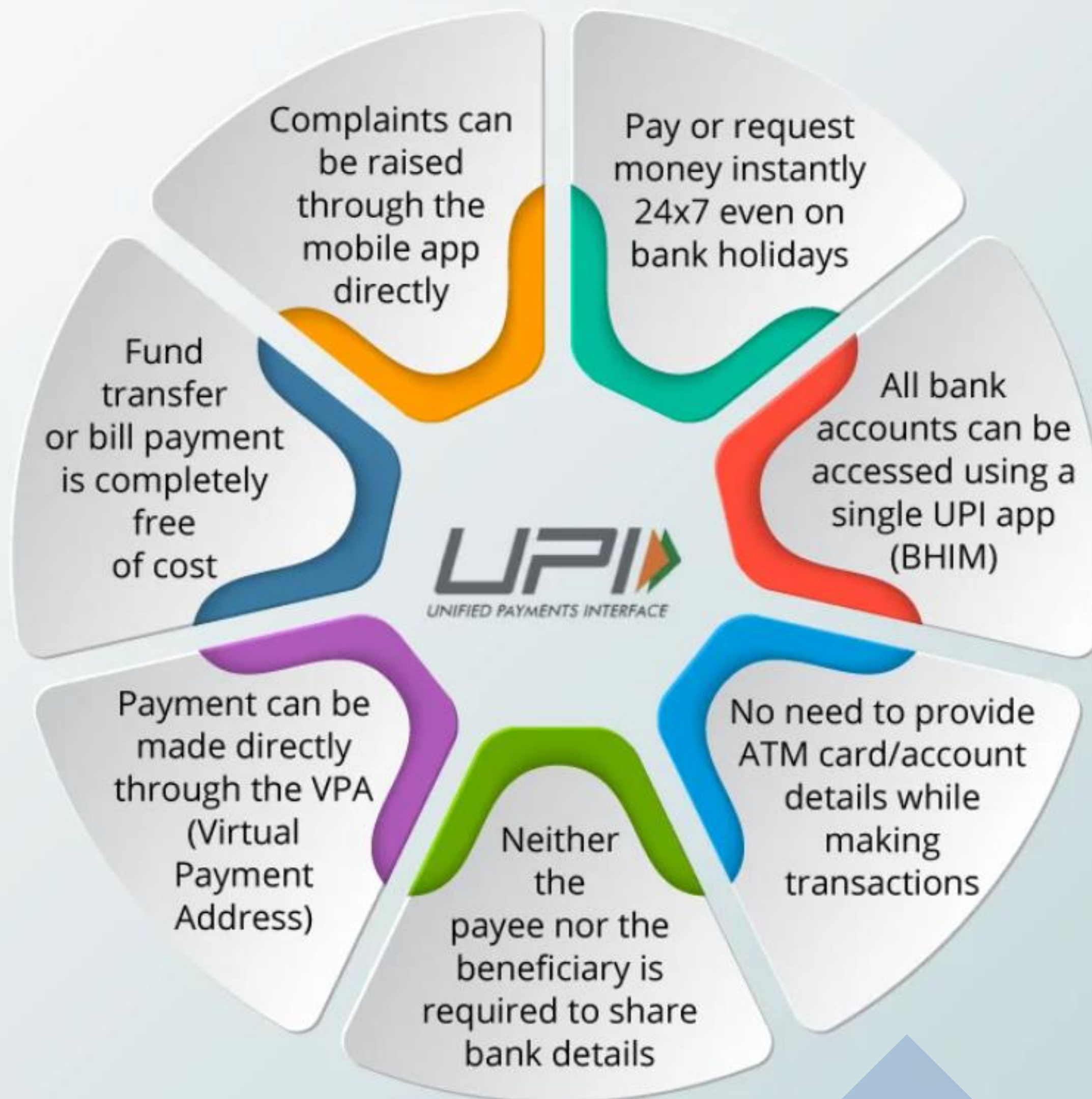


Working of Card Based Payment Systems





- **UPI:** The Unified Payments Interface (UPI) is a revolutionary payment system that has transformed the financial landscape of India. Developed by the National Payments Corporation of India (NPCI) and regulated by the Reserve Bank of India (RBI), UPI allows users to transfer funds instantly and seamlessly between bank accounts using a mobile app.





- **Key Features:**

- **Real-time:** Transactions are completed instantly, 24/7, eliminating the need to wait for funds to clear.
- **Interoperable:** Works across different banks, enabling payments between any two UPI-enabled accounts.
- **Virtual Payment Address (VPA):** Unique identifier for each user, eliminating the need to share bank account details.
- **Mobile-based:** Requires only a smartphone and internet connection, making it accessible to a wider population.
- **Secure:** Uses multiple layers of security, including PIN authentication and end-to-end encryption.
- **Cost-effective:** Most transactions are free of charge, promoting financial inclusion.



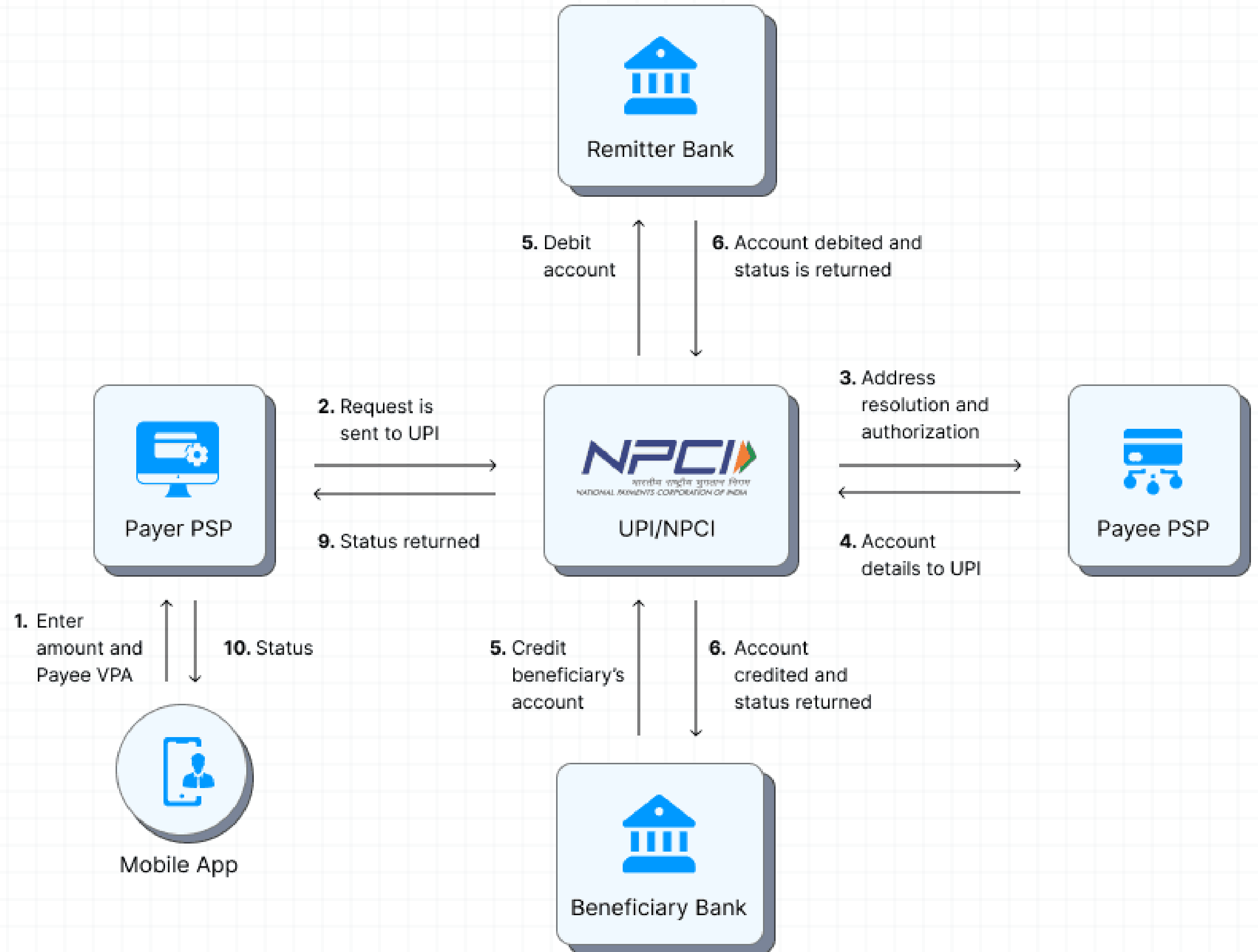
- **Types of UPI Transactions:**
 - **Peer-to-Peer (P2P):** Transfer funds between individuals using their VPA or mobile number.
 - **Peer-to-Merchant (P2M):** Pay bills and make purchases at online and offline stores.
 - **Bill Payments:** Pay utility bills, insurance premiums, and other recurring bills.
 - **Collect Request:** Request money from another user by sending a virtual payment request.
 - **Split Bills:** Easily split bills and expenses among friends and colleagues.



- **UPI Terminology:**

- **VPA (Virtual Payment Address) or UPI ID:** It's similar to an email id and consists of the issuing bank and a unique identifier.
- **Issuer PSP:** The Bank or PSP(Payment Service Provider) which creates the VPA.
- **Acquirer PSP:** The Bank or PSP which processes VPA transactions.
- **Payee:** Person/entity which is receiving the money.
- **Payer:** Person/entity who is paying the money.

- Working of UPI:





- **UPI Transaction Limits:**

- The maximum UPI limit per day for normal transactions is Rs. 1 Lakh
- The UPI limit per day for ATM withdrawal using UPI app is Rs. 10,000
- A total of 20 UPI transactions can be done through bank UPI apps per day
- One can carry out a maximum of 10 transactions per day through third-party UPI apps.
- Per transaction limit for every UPI transaction is Rs. 1 Lakh
- For transactions related to capital markets, collections, insurance, and foreign inward remittances, the UPI limit is Rs. 2 Lakh per transaction
- The UPI limit for IPOs and retail direct scheme is Rs. 5 Lakh per transaction



- **Benefits for Users:**
 - Highly secure as no account detail is shared
 - VPA ensures no personal details are shared with others
 - Round the clock transaction facility
 - Easy to use
 - The transfer can be made instantly
 - All bank accounts can be managed through a single app
 - Single click authentication
 - Money can be requested from the payee through the app
 - Complaints can be raised directly as well

QR Code (Quick Response Code):

- Quick Response (QR) codes are two-dimensional barcodes that can store a variety of information, including URLs, text, contact details, and even images.
- They are readily recognizable by their square shape and black and white squares arranged in a grid pattern.
- QR codes were invented in 1994 by Denso Wave, a Japanese subsidiary of Toyota, to track vehicles during manufacturing. However, they quickly gained popularity for a variety of applications due to their ability to store large amounts of information in a small space.
- QR codes are encoded using a specific algorithm that translates data into a pattern of black and white modules.
- The code can then be scanned by a smartphone or dedicated QR code reader, which decodes the data and displays it on the screen.



Types of QR Codes:

- **URL QR codes:** Store a website address, allowing users to access it directly by scanning the code.
- **Text QR codes:** Store plain text information, such as contact details or instructions.
- **vCard QR codes:** Store contact information, which can be automatically added to the user's address book.
- **Email QR codes:** Pre-populate email addresses and subject lines, allowing users to send emails directly from the QR code.
- **SMS QR codes:** Pre-populate text messages, allowing users to send messages directly from the QR code.
- **Wi-Fi QR codes:** Store Wi-Fi network credentials, allowing users to connect to a network automatically.



Benefits of QR Codes:

- **Convenience:** QR codes offer a quick and easy way to access information, eliminating the need to manually type URLs or contact details.
- **Accessibility:** QR codes are accessible to anyone with a smartphone and a QR code reader app, making them a versatile tool for a wide range of applications.
- **Versatility:** QR codes can store a wide variety of information, making them suitable for diverse applications such as marketing, advertising, product packaging, ticketing, and even personal use.
- **Security:** QR codes can be encrypted to ensure the security and privacy of the stored information.
- **Cost-effective:** QR codes are relatively inexpensive to create and use, making them a suitable option for businesses and individuals alike.



National Relief Fund
Account No: 007010100273206
IFSC: UTIB0000007

SCAN & PAY

Applications of QR Codes:





UPI
UNIFIED PAYMENTS INTERFACE



National Relief Fund
Account No: 007010100273206
IFSC: UTIB0000007

SCAN & PAY

Select your preferred language

☐ Hello (ENGLISH)

☐ नमस्ते (HINDI)

☐ নমস্কার (BENGALI)

☐ నమస్కారం (TELUGU)

☐ नमस्ते (MARATHI)

☐ வணக்கம் (TAMIL)

☐ اداپ (URDU)

☐ ನಮಸ್ಕಾರ (KANNADA)



User Name

9999999999@upi

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SERVICES



Mandates



Tap



Scan



Voice



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RECOMMENDED



User Name

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9999999999@upi

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MY LINKED BANK ACCOUNTS



ABC Bank
XXX 1736



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OTHER ACCOUNTS



UPI LITE Balance

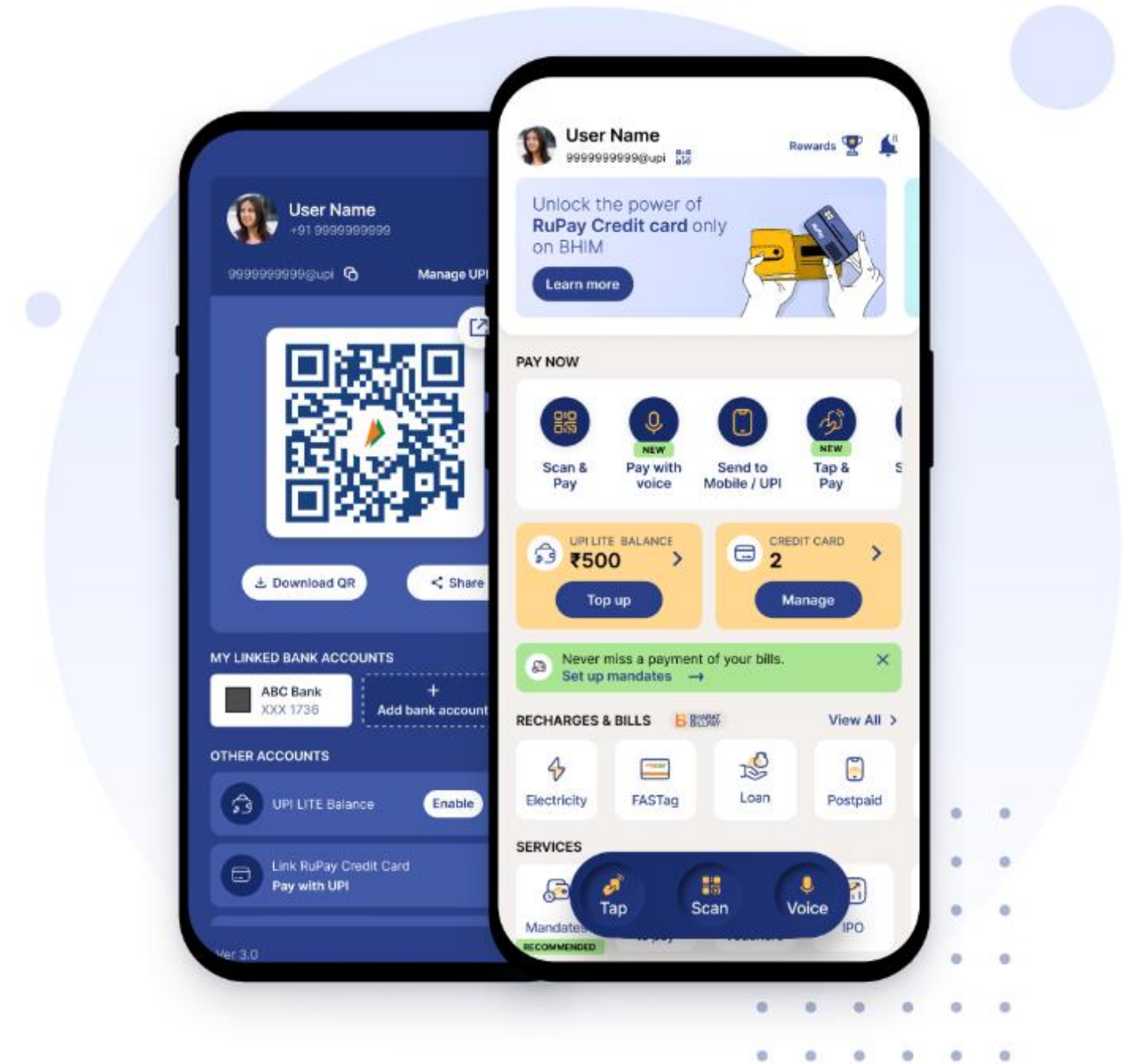
[Enable](#)



[Link RuPay Credit Card
Pay with UPI](#)

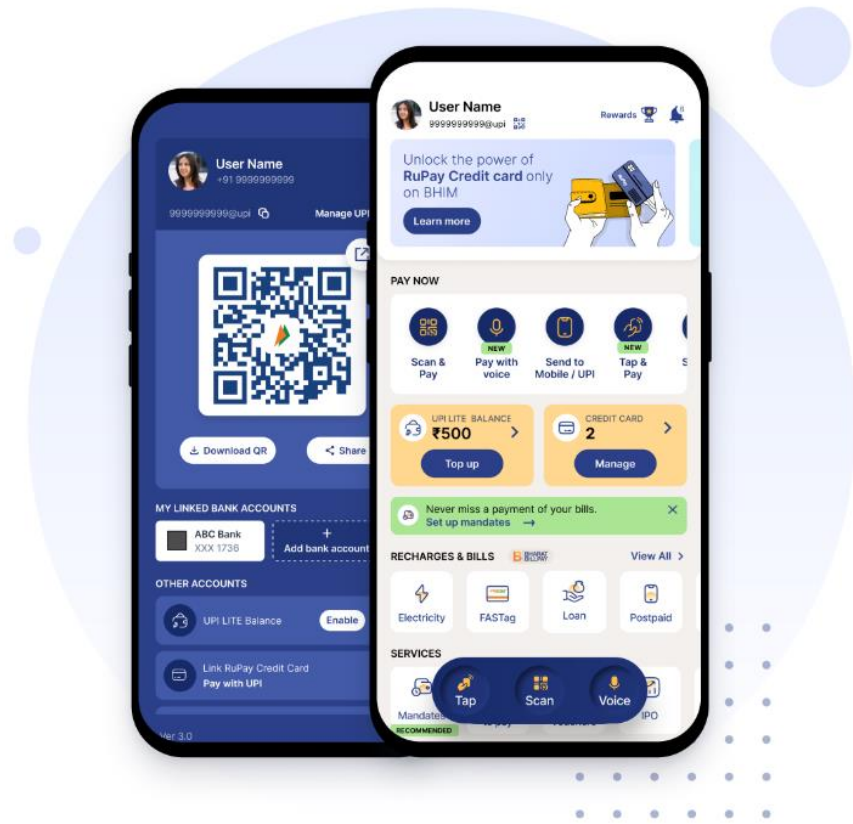
Ver 3.0

- **BHIM (Bharat Interface for Money):**
 - BHIM, also known as Bharat Interface for Money, is a mobile payment app launched by the National Payments Corporation of India (NPCI) in December 2016.
 - It is based on the Unified Payments Interface (UPI) and facilitates instant and secure digital payments between individuals and businesses.
 - You can make direct bank payments to anyone on UPI using their UPI ID or scanning their QR with the BHIM app.
 - You can also request money through the app from a UPI ID.



Benefits:

- **UPI-based:** Leverages the power of UPI to offer real-time money transfers using a mobile app.
- **Interoperable:** Works across different banks, enabling payments between any two UPI-enabled accounts.
- **Virtual Payment Address (VPA):** Unique identifier for each user, eliminating the need to share bank account details.
- **Multiple payment options:** Supports various payment methods, including transferring funds, paying bills, and splitting bills.
- **Secure:** Utilizes multiple security features, including PIN authentication and end-to-end encryption.
- **Simple and user-friendly:** Designed for ease of use, making it accessible to users of all technical backgrounds.
- **Cost-effective:** Most transactions are free of charge, promoting financial inclusion.



USSD



***99#**



Dial *99# on
your phone
and press the
call button.



USSD

USSD (Unstructured Supplementary Service Data):

- **Unstructured Supplementary Service Data (USSD)** is a technology used to send and receive text messages between a mobile phone and a network application.
- It is a part of the GSM standard and is supported by most mobile phones worldwide.
- One more step towards reaching the last mile of the disconnected part of our society and bringing them into the growing financial inclusion and mainstream banking and enabling them to transact digitally, the National Payment Corporation of India (NPCI) launched *99#.
- *99# is an Unstructured Supplementary Structured Data (USSD) based mobile banking service that enables banking customers to access its various key services without using data services or the internet.



USSD

Working of USSD:

- USSD messages are initiated by dialing a specific code on the mobile phone.
- This code typically starts with an asterisk (*) and ends with a hash (#).
- The code is then sent to the network application, which processes the request and sends a response back to the mobile phone as a text message.
- USSD messages are limited to 182 characters.
- They can be used to access a variety of services, including checking account balances, transferring funds, and purchasing goods and services.



USSD

Features of *99# USSD Service:

- Uses USSD as the access channel that works across all GSM handsets (smartphone or otherwise) making it reach the last mile user
- Supports menu-based applications that is easy to maneuver for the users
- Does not require data connectivity (works on signaling channel) that makes it high availability service
- Round the clock availability (works even on holidays)
- Accessible through a common code *99# across all GSM Operators and mobile handsets
- Additional channel for using BHIM app and key catalyst for financial inclusion



USSD

Steps to use *99# USSD Facility:

- Link your mobile number to your bank account
- Dial *99# from your phone
- Fill in the first 3 letters of your bank against short name or first 4 letters of IFSC
- Choose “Fund-Transfer-MMID” option
- Enter the payee’s mobile number and MMID
- Enter the amount and your MPIN, leave a space and enter the last 4 digits of your account number.



USSD

Benefits of USSD:

- **Accessibility:** USSD is available on most mobile phones, making it a widely accessible technology.
- **Simple to use:** USSD is easy to use and requires no special knowledge or equipment.
- **Cost-effective:** USSD messages are typically free or very cheap to send and receive.
- **Offline functionality:** USSD can be used even when the mobile phone is not connected to the internet.



USSD

Limitations of USSD:

- **Limited data transfer:** USSD messages are limited to 182 characters, which can be restrictive for some applications.
- **No multimedia support:** USSD cannot be used to send or receive multimedia content, such as images or videos.
- **Session-based:** USSD sessions are limited in time and require the user to re-enter the code if they want to continue using the service.
- **Security concerns:** USSD messages are not encrypted by default, which can be a security risk in some cases.



USSD

Applications of USSD:

- **Banking:** Checking account balances, transferring funds, paying bills.
- **Mobile money:** Sending and receiving money, purchasing goods and services.
- **Information services:** Checking weather forecasts, news updates, sports scores.
- **Government services:** Applying for licenses, paying taxes, registering for elections.
- **Customer support:** Contacting customer service representatives, getting help with products and services.





Cash withdrawal

Mini Statement





- AEPS is a bank-led model that utilizes the unique biometric authentication of Aadhaar to enable financial transactions.
- Users can access their bank accounts and perform various financial services using their Aadhaar number and fingerprint authentication through micro-ATMs or Point of Sale (PoS) terminals.
- It allows bank-to-bank transaction at PoS (Micro ATM) with the help of Banking Correspondent (BC)/ **Village Level Entrepreneur (VLE)**.
- The user must seed his/her account with their Aadhaar number at bank or with the help of BC/VLE.
- User can do as many transactions as possible at any AEPS point without any PIN or Password.



Key Features of AEPS:

- **Biometric authentication:** Ensures secure and reliable transactions using fingerprint verification.
- **Interoperable:** Works across all banks participating in the AEPS network.
- **Real-time transactions:** Provides instant access to funds and eliminates waiting periods.
- **Accessibility:** Requires only an Aadhaar number and fingerprint, making it accessible even to those without smartphones or internet access.
- **Cost-effective:** Offers low transaction fees compared to traditional banking methods.



Functionalities of AEPS:

- **Cash withdrawal:** Users can withdraw cash from their bank accounts using micro-ATMs or PoS terminals.
- **Balance inquiry:** Users can check their account balance instantly.
- **Cash deposit:** Users can deposit cash into their bank accounts through authorized agents.
- **Fund transfer:** Users can transfer funds between their own accounts or to other Aadhaar-linked accounts.
- **Merchant payments:** Users can make payments to merchants for goods and services using their Aadhaar number.



Steps to use AEPS:

- 1 Go to a Micro ATM or Banking Correspondent.
- 2 Provide Your Bank Name and Aadhar.
- 3 Choose kind of transaction to do.
- 4 Provide Fingerprint on scanner.
- 5 Take the printed slip after successful transaction.



Impact of AEPS:

- **Promoting financial inclusion:** AEPS has brought millions of unbanked and underbanked individuals into the formal financial system, boosting financial literacy and economic participation.
- **Enhancing financial security:** Biometric authentication reduces the risk of fraud and identity theft, creating a more secure financial environment.
- **Improving financial access:** AEPS provides convenient and affordable access to financial services, particularly for those residing in rural areas or with limited internet access.
- **Boosting digital payments:** AEPS has fueled the growth of digital payments in India, promoting a cashless economy and increasing transparency in financial transactions.



E-Wallet



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E-Wallets:

- E-wallet, Digital Wallet or Mobile wallet is a digital account, which is equivalent to a physical wallet.
- It is a digital application or platform that stores your financial information and allows you to conduct various financial transactions without needing physical cash or cards.
- You can link your bank accounts, credit cards, or debit cards to your e-wallet and use it to pay your utility bills, recharge mobile and DTH accounts, book train, bus or airline tickets, transfer money to another wallet or bank account, get cash from agents and retailers, apart from online as well as offline shopping.



Key features of e-wallets:

- **Convenience:** E-wallets eliminate the need to carry cash or cards, making transactions faster and easier.
- **Security:** E-wallets are protected by multiple security layers, including passwords, PINs, and biometric authentication.
- **Contactless payments:** E-wallets enable contactless transactions through NFC technology, reducing the risk of contact-based infections.
- **Multi-functionality:** Many e-wallets offer additional features like bill payments, loyalty programs, and budgeting tools.
- **Accessibility:** E-wallets are readily available on smartphones and can be used by anyone with a mobile device.



Types of e-wallets:

- **Open wallets:** These wallets allow users to store funds and conduct various transactions, including payments, transfers, and bill payments.
- **Closed wallets:** These wallets are issued by specific merchants or companies and can only be used for transactions within their ecosystem.
- **Semi-closed wallets:** These wallets offer a combination of open and closed functionalities, allowing users to store funds and make payments within a specific network but also accept funds from other wallets.



Benefits of using e-wallets:

- **Increased efficiency:** Transactions are faster and easier compared to using cash or cards.
- **Reduced transaction costs:** E-wallets often offer lower transaction fees than traditional methods.
- **Enhanced financial control:** E-wallets provide real-time transaction tracking and budgeting tools.
- **Improved security:** Biometric authentication and encryption features protect users from fraud and identity theft.
- **Greater accessibility:** E-wallets offer financial inclusion to individuals who may not have access to traditional banking services.

A word cloud featuring various question words in different sizes and orientations. The word 'WHAT?' is the largest and most prominent, located in the center. Other large words include 'WHY?', 'WHERE?', 'WHEN?', 'HOW?', and 'WHO?'. Smaller words like 'WHICH?', 'WHOSE?', 'WHAT?', 'HOW?', 'WHY?', 'WHERE?', 'WHEN?', 'WHO?', and 'WHICH?' are scattered around the larger words, some appearing multiple times. The words are arranged in a way that they seem to be floating or falling, creating a dynamic and abstract composition.

WHAT? HOW? WHY?
 WHO? WHERE? WHAT? HOW?
 WHERE? WHICH? WHOSE? WHEN? WHY?
 WHY? HOW? WHERE?
 WHO? WHOSE?
 WHERE? WHAT? HOW?
 WHERE? WHICH? WHOSE? WHEN? WHY?
 WHAT?
 WHO?
 WHY?
 HOW?
 WHERE?
 WHEN?
 WHOSE?

A word cloud featuring the five Ws and H (Who, What, Where, Why, How) in various sizes and orientations. The words are arranged in a circular pattern, with 'WHERE?' and 'HOW?' being the largest and most prominent. Other words like 'WHO?', 'WHAT?', and 'WHY?' are also visible in smaller sizes. The background is a solid light blue.

